

Methodology, Experimental Framework, and Results

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1 Methodology

1.1 Problem Formulation and Design Objectives

Given pre-event imagery I^{pre} and one or more post-event observations I_t^{post} , we seek a set of persistent building records rather than independent image-level predictions. For building i , the desired record links its geometry g_i to a building-type distribution $\hat{\tau}_i$, an exposure proxy $\hat{p}_i$, date-specific ordinal damage estimates $\hat{y}_{i,t}$, and fields describing observation coverage and indicators of predictive uncertainty. This formulation makes the mapped building, rather than the image tile, the unit through which evidence is accumulated and delivered.

The formulation serves three design objectives. First, spatial identity must persist across delineation, image extraction, model inference, temporal aggregation, and presentation. Second, the semantics of each estimate must remain explicit: the exposure value is a constructed proxy, ordinal damage is a model inference, and an absent observation is not a negative finding. Third, intermediate coverage and uncertainty indicators must survive synthesis so that decision makers can distinguish mapped evidence from missing or ambiguous evidence. The objective is therefore not a single impact score, but an inspectable geographic record that can support filtering, prioritization, and visual review.

1.2 Instance-Centric GeoAI Framework

The framework connects four named stages through a shared building-instance contract (Figure 1). Stage 1, *Building-Instance Delineation and Asset Generation*, extracts candidate geometries from pre-event imagery and creates aligned crops, masks, and metadata. Stage 2a, *Building-Type and Exposure-Proxy Estimation*, and Stage 2b, *Ordinal Building-Damage Estimation*, operate as parallel inference branches over those shared assets. Stage 3, *Coverage-Aware Multi-Date Synthesis*, then combines the available outputs across models and dates. Each stage can be revised independently because its inputs and outputs refer to the same geographic object.

A canonical identifier is propagated with the geometry and source-tile metadata. Input adapters normalize module-specific field names before inference, and synthesis forms the union of identifiers rather than an inner join. Stage-presence flags consequently distinguish a missing prediction from a negative prediction. The same contract also retains confidence, uncertainty indicators such as entropy and between-model disagreement, source paths, and valid-observation counts. Table 1 summarizes how these fields change the interpretation of the final information product.

Because the exposure variable is constructed rather than directly observed, we define its data lineage before returning to the operational order from Stage 1 through Stage 3.

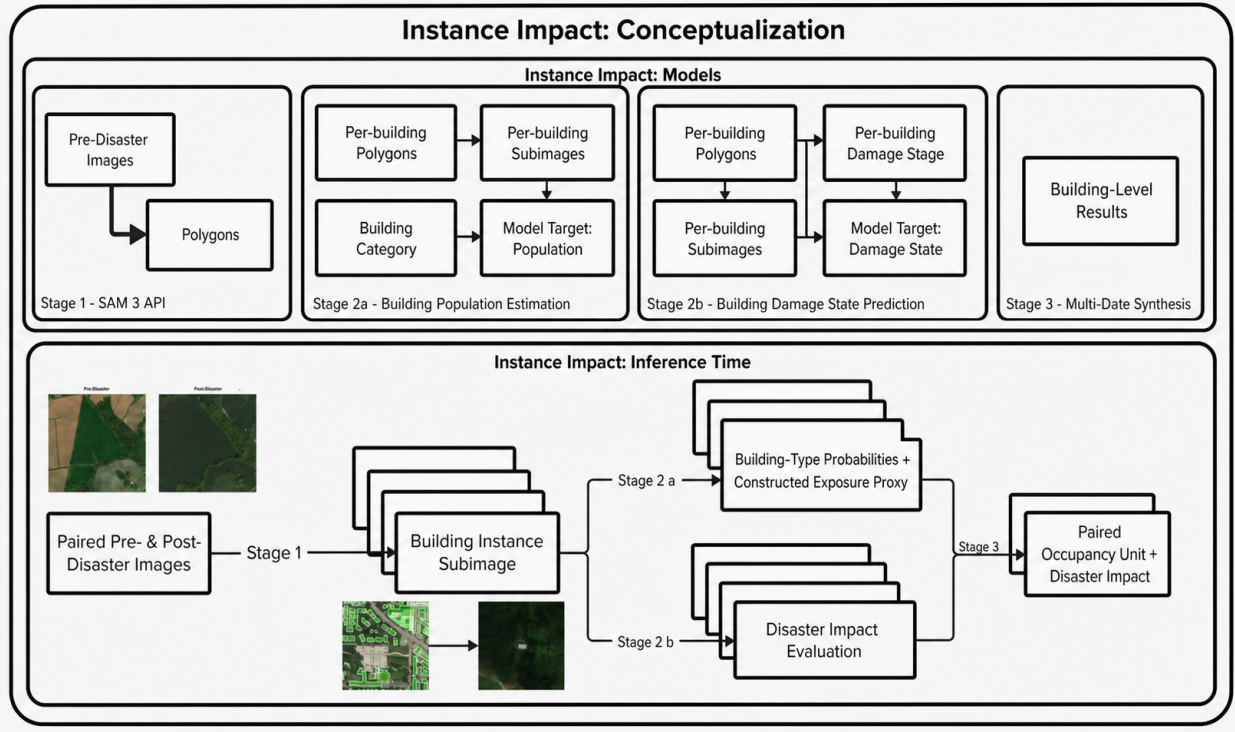


Figure 1: Conceptual organization of Instance Impact. The upper band associates the four stages with their model outputs, while the lower band follows persistent building instances through the parallel Stage 2a exposure-proxy and Stage 2b damage branches to Stage 3 synthesis. Population and occupancy wording in the schematic denotes a constructed exposure proxy rather than observed occupants; the final product is a building-level evidence record rather than a calibrated impact score.

Table 1: Decision-facing semantics of the building-instance contract.

Information element	Encoded distinction	Decision-facing use
Persistent identifier	Same building across crops, models, and dates	Prevents cross-stage record ambiguity
Stage-presence flags	Missing prediction versus negative prediction	Makes model coverage filterable and countable
Target lineage	Constructed exposure versus observed population	Bounds interpretation of the exposure estimate
Validation-source field	Parcel code versus area heuristic in labelled data	Supports source-stratified evaluation
Damage class –1	Not identifiable versus no visible damage	Prevents invalid imagery from implying safety
Entropy and disagreement	Lower versus higher uncertainty indicators	Directs ambiguous instances to visual review

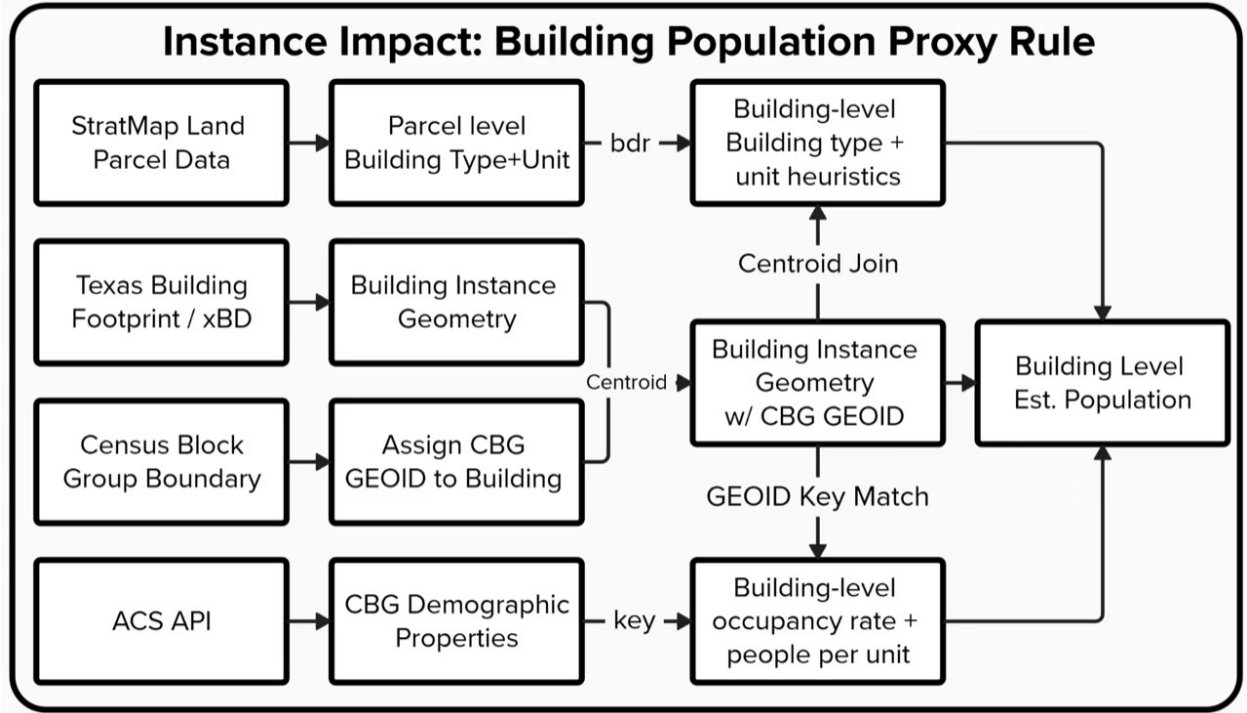


Figure 2: Workflow used to construct the building-level exposure-proxy target. Parcel records provide building-type and unit heuristics; xBD footprints supply building geometry; and CBG boundaries link each building to ACS occupancy and people-per-unit attributes. Their building-level combination produces the Stage 2a training target, which is treated throughout this paper as a constructed exposure proxy rather than an observed count of occupants.

1.3 Stage 2a: Exposure-Proxy Data and Target Construction

The exposure-training data combine xBD Hurricane Harvey pre-disaster imagery and building polygons (Gupta et al., 2019), Harris County parcel land-use records, 2022 TIGER/Line Census Block Group (CBG) boundaries (U.S. Census Bureau, 2022), and 2018–2022 American Community Survey five-year estimates (U.S. Census Bureau, 2023) (Figure 2). The xBD labels provide pixel- and geographic-coordinate polygons and image ground-sample distance. We compute each footprint area from its pixel polygon and squared ground-sample distance, and use its geographic centroid for the demographic and parcel joins.

Each polygon is expanded by a 20m buffer, clipped to its 1024×1024 source tile, and saved as an RGB crop with a binary footprint mask. CBG and parcel layers are reprojected to WGS84 before point-in-polygon joins. Duplicate spatial matches are reduced to one row per building. The resulting label table contains 23,014 buildings from 300 image tiles, of which 21,140 have a CBG identifier. Parcel codes provide 4,554 building-type assignments, while 18,460 are obtained from footprint-area rules.

Parcel assignments are accepted only when their reported use is plausible for the footprint size; otherwise the area rule is used. Single-family and mobile-home buildings are assigned one unit, duplexes two, and triplexes three. For multi-family buildings with footprint area a_i , the unit estimate is capped at 40:

Table 2: Building-type distribution before target-category selection.

Class	Buildings
Commercial	17,392
Residential multi	3,687
Other	954
Institutional	632
Residential small	349

$$u_i = \begin{cases} \min\{40, \max(1, \lfloor a_i/90 \rfloor)\}, & a_i < 2,000 \text{ m}^2, \\ \min\{40, \max(1, \lfloor a_i/80 \rfloor)\}, & a_i \geq 2,000 \text{ m}^2. \end{cases} \quad (1)$$

Detailed parcel and area-rule types initially yield five model classes (Table 2). Institutional combines educational, religious, charitable, exempt, and miscellaneous uses, while other combines industrial, agricultural, oil and gas, utility, vacant, and unknown uses. These two categories therefore mix buildings with incompatible occupancy processes. We exclude the 632 institutional and 954 other instances before splitting or fitting. The training dataset consequently contains 21,428 instances across commercial, residential multi, and residential small buildings. This exclusion is a limitation: the training domain does not represent institutional and other buildings.

For residential building i in CBG g , the exposure proxy is constructed as

$$\tilde{p}_i = \min(c_{t_i}, u_i \rho_{g,t_i} o_g), \quad (2)$$

where u_i is the estimated number of units, ρ_{g,t_i} is the CBG-specific people-per-unit ratio for type t_i , o_g is the CBG occupancy rate, and c_{t_i} is a type-specific cap. Missing CBG ratios use type-specific defaults of 2.85, 2.40, 2.50, 2.20, 2.00, 1.70, and 2.40 people per unit for single family, mobile home, duplex, triplex, small multi-family, large multi-family, and unknown residential, respectively; the default occupancy rate is 0.91. The corresponding caps are 12, 8, 20, 30, 120, 500, and 12 people.

For commercial buildings, the construction applies a daytime density to footprint area,

$$\tilde{p}_i = a_i d_{t_i}, \quad (3)$$

with $d_{t_i} = 0.040$ people per square metre. Thus $\tilde{p}_i$ is a reproducible exposure proxy assembled from cadastral, areal-demographic, and geometric rules; it is not a direct observation of building occupancy. The recorded construction source is used for stratified evaluation but is not required at inference.

1.4 Stage 1: Building-Instance Delineation and Asset Generation

Building instances are delineated from pre-event imagery so that damage does not redefine the asset boundary. SAM 3 (Carion et al., 2025) is prompted with *building*; each positive instance mask is polygonized separately and associated with its mean segmentation score. The resulting geometry, score, source image, and identifier establish the spatial anchor for all later estimates.

For each polygon, the asset generator creates aligned 256×256 pre- and post-event RGB crops, a target-footprint mask M , and a 48-pixel context-ring mask R . It also records crop coordinates, event and hazard metadata, footprint and ring areas, source paths, and delineation confidence. Both branches read this shared instance package but select branch-specific inputs: Stage 2a uses the pre-event crop, footprint mask, and geometry, whereas Stage 2b uses paired pre- and post-event



Figure 3: Illustrative Stage 1 building-instance delineation for Los Angeles cell 00507. (a) Pre-event imagery used to define persistent building geometry. (b) In-frame boundaries from the current packaged Stage 1 output, with an enlarged view of individual instances. This application example shows the spatial interface passed downstream; it is not part of the xView2 benchmark used to compute the delineation metrics in Section 3.1.

crops with footprint and context masks. This shared representation ensures that type, exposure, and damage estimates refer to the same geometry rather than independently reconstructed versions of a building.

Figure 3 illustrates the spatial object produced by Stage 1 for a packaged application cell. The delineated boundaries remain linked to the pre-event image and provide the geometry from which the shared downstream assets are generated.

1.5 Stage 2a: Building-Type and Exposure-Proxy Estimation

Stage 2a separates visual building-type inference from exposure-proxy regression (Figure 4a). Each type classifier receives a 224×224 pre-event RGB crop with the footprint mask appended as a fourth channel. The same mask is downsampled to the final ConvNeXt-Tiny feature map F_i and used for target-specific average pooling (Liu et al., 2022):

$$\mathbf{h}_i^M = \frac{\sum_{x,y} F_i(x,y) M_i(x,y)}{\max(\varepsilon, \sum_{x,y} M_i(x,y))}. \quad (4)$$

The 768-dimensional pooled feature is projected to 512 dimensions. Four deployable geometric attributes (log footprint area, mask fill ratio, bounding-box aspect ratio, and compactness) are projected from four to 64 dimensions. Their concatenation passes through a $576 \rightarrow 128 \rightarrow 3$ head that estimates commercial, residential-multi, and residential-small probabilities. Five independently fitted classifiers produce probability vectors $\mathbf{q}_i^{(m)}$, which are averaged at inference:

$$\hat{\tau}_i = \frac{1}{5} \sum_{m=1}^5 \mathbf{q}_i^{(m)}. \quad (5)$$

The exposure model uses predicted rather than reference type. A standardized ridge model estimates a baseline b_i for $\log(1 + \tilde{p}_i)$ from log footprint area. An absolute-error histogram gradient-boosting model estimates a residual r_i from log footprint area, the three soft type probabilities, their maximum, probability margin, entropy, and the combined residential probability. The final estimate is

$$\hat{p}_i = \max\{0, \exp(b_i + r_i) - 1\}. \quad (6)$$

This staged construction preserves the area-only baseline while allowing visual building-use evidence to explain systematic residual variation. It also prevents the constructed exposure target from being presented as a direct image-derived occupancy count.

The model uses footprint area supplied by the instance record. If upstream area is unavailable, the inference adapter approximates it from footprint-mask pixels as

$$\hat{a}_i = 4.641392 A_i^{\text{mask}}, \quad (7)$$

where A_i^{mask} is mask area in pixels. Confidence, margin, entropy, soft class probabilities, and ensemble variation remain attached to each prediction. The exposure estimate can therefore be reviewed together with the type evidence that informed it.

1.6 Stage 2b: Ordinal Building-Damage Estimation

Damage is represented by four ordered states: no visible damage, minor damage, major damage, and destroyed. A weight-shared ConvNeXt-Tiny backbone processes the 256×256 pre- and post-event RGB crops (Figure 4b). The final feature maps are pooled separately over the footprint M and context ring R . For region $S \in \{M, R\}$, the signed change is $\Delta_i^S = \mathbf{h}_{i,\text{post}}^S - \mathbf{h}_{i,\text{pre}}^S$. The fused representation is

$$\mathbf{z}_i = [\mathbf{h}_{i,\text{pre}}^M, \mathbf{h}_{i,\text{post}}^M, \mathbf{h}_{i,\text{pre}}^R, \mathbf{h}_{i,\text{post}}^R, \Delta_i^M, \Delta_i^R, |\Delta_i^M|, |\Delta_i^R|]. \quad (8)$$

The final ConvNeXt stage has $C = 768$ channels, giving an $8C = 6,144$ dimensional vector. A $6,144 \rightarrow 512 \rightarrow 3$ head with ReLU activation and dropout produces cumulative logits for the ordered targets $y > 0$, $y > 1$, and $y > 2$. This CORAL-style formulation (Cao et al., 2020) encodes ordered thresholds through its training targets. Adjacent sigmoid differences are clamped and normalized to obtain four class probabilities.

Inference combines three checkpoints with weights 4:3:2. Weighted raw cumulative logits determine the damage class. Temperature-scaled member probabilities provide expected severity, confidence, entropy, and between-model disagreement. These quantities remain review fields rather than being collapsed into the ordinal label.

1.7 Stage 3: Coverage-Aware Multi-Date Synthesis

Synthesis creates one row for every identifier present in any stage. The record includes geometry, delineation confidence, type and exposure estimates, ordinal damage, expected severity, uncertainty indicators, source paths, and stage-presence flags. The union join makes partial coverage explicit and permits analysts to filter buildings by which evidence is available.

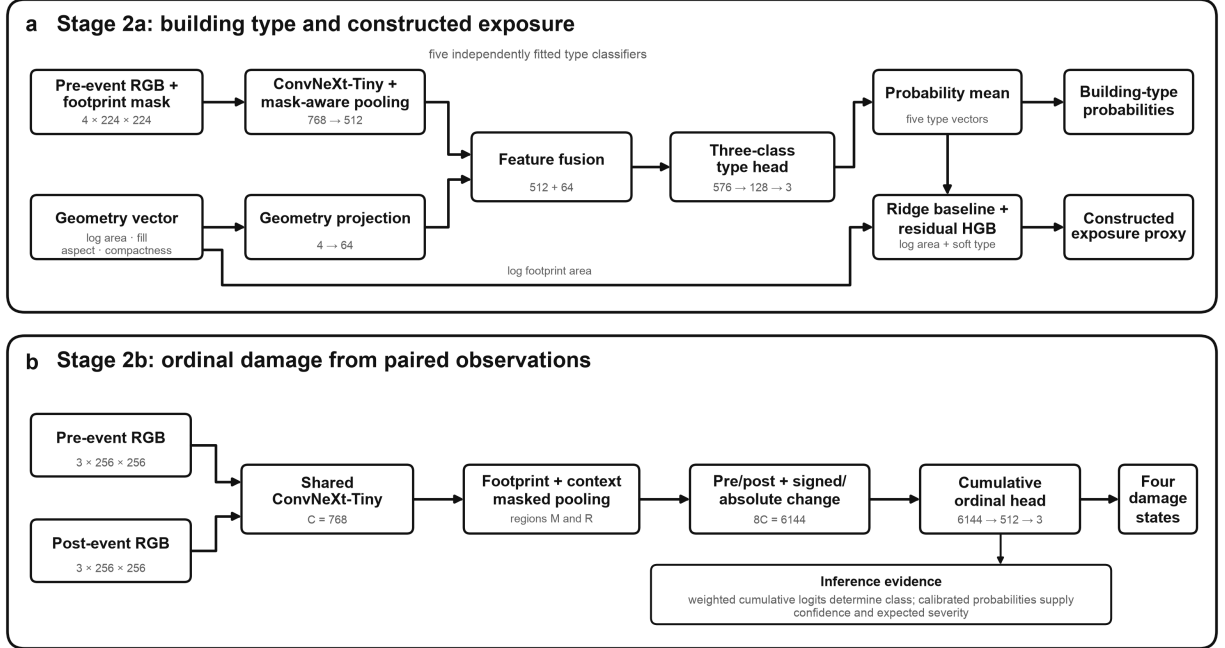


Figure 4: Architectures of the two learned Stage 2 branches. (a) Stage 2a combines mask-aware pre-event image features with deployable geometry to estimate three building-type probabilities. Their five-model mean informs a separately fitted ridge-plus-residual exposure model. (b) Stage 2b applies a weight-shared backbone to paired observations, pools footprint and context features, and combines state and change evidence in a cumulative ordinal head.

For repeated post-event observations, a date contributes to the building-level damage estimate only when both tile quality and crop coverage are valid. The final class is the majority among valid dates, with ties resolved toward the more severe class. A building with no valid date receives class -1 (not identifiable), rather than class 0 (no visible damage). Temporal coverage therefore remains part of the substantive result instead of being treated as a preprocessing detail.

For optional review ordering, the system computes

$$s_i = \hat{p}_i \mathbb{E}[y_i], \quad (9)$$

where $\hat{p}_i$ is the exposure proxy and $\mathbb{E}[y_i]$ is expected ordinal severity. This product is used only as a heuristic ranking signal, not a calibrated impact score. The deliverables remain the full instance table, coverage summary, uncertainty queue, and spatial overlays, allowing a reviewer to recover the evidence behind any prioritized building.

2 Experimental Framework

2.1 Study Design and Geographic Evidence

The experiments follow the information pathway from building discovery to multi-date delivery. Each evaluation is tied to the geographic unit and inferential role of its component (Table 3). Polygon matching measures whether buildings enter the information system; tile-blocked validation evaluates exposure-proxy and damage models without sharing source tiles across partitions;

component comparisons isolate the contributions of footprint masks, type probabilities, and neighbourhood context; and the Los Angeles application examines how predictions and unknown states are assembled across repeated observations.

Table 3: Geographic domains, evaluation units, and evidential roles.

Component	Domain and unit	Evaluation design	Evidential role
Building delineation	xView2 multi-disaster test set; 933 images	One-to-one polygon matching at IoU 0.5	Measures instance discovery and matched geometry
Type and exposure	Harris County, Hurricane Harvey; 21,428 buildings	Tile-blocked validation, repeated training, ablation, and alternative partitions	Measures proxy reconstruction and geographic/source sensitivity
Ordinal damage	xBD flooding events; 41,582 buildings	Matched tile-blocked context comparisons and repeated training	Measures the contribution of target-neighbourhood context
Multi-date synthesis	Los Angeles wildfire; 21,797 buildings and five dates	Cross-hazard application without wildfire-specific retraining	Examines explicit unknown states and analyst-facing delivery

The component evaluations and the application serve different purposes. The xView2 and xBD experiments quantify delineation, proxy reconstruction, and ordinal damage within their respective evaluation domains. The Los Angeles case is a cross-hazard demonstration of information handling: it evaluates whether the instance contract preserves temporal coverage and produces reviewable outputs, rather than estimating wildfire-damage accuracy.

2.2 Model Training and Validation

Training and validation preserve the building instance as the sampling unit while assigning every instance from a source tile to a single partition. Table 4 records the settings needed to reproduce the learned Stage 2 branches; the following text describes how these settings preserve the relationship between model fitting and deployment.

Building delineation and asset retention. The delineation evaluation uses 933 xView2 test images and one-to-one matching between predicted and reference polygons at IoU 0.5.

Building type and exposure proxy. The 21,428 labelled buildings are grouped by source tile. Five type classifiers are fitted independently on the common training partition with weighted sampling and inverse-square-root class weights. Training augmentation synchronizes horizontal flips and right-angle rotations across each crop and its footprint mask. Exponential moving-average weights are evaluated after each epoch, and the checkpoint with the highest validation macro-F1 is retained.

The exposure regressor is trained only after the type models. Within each type-model seed, five tile-blocked folds generate out-of-fold probabilities for all training buildings. This prevents the exposure regressor from receiving type probabilities produced by a classifier that saw the same building during fitting. The final regressor uses probabilities averaged across the out-of-fold seed families, while validation and inference use the mean of the five full-training classifiers. This protocol aligns the training-time type features with the probability ensemble available at deployment.

Ordinal damage. The flood-damage dataset contains 41,582 buildings: 31,225 with no visible damage, 5,134 minor, 4,721 major, and 502 destroyed. Whole image tiles are assigned to training or a 15% validation partition. Training applies synchronized horizontal flips and right-angle rotations

Table 4: Training, selection, and inference protocols for the learned Stage 2 components.

Component	Geographic protocol	Fitting objective and optimization	Selection and inference
Stage 2a type	18,224 training and 3,204 validation buildings; fixed tile-blocked split; five seeds	Weighted cross-entropy; ConvNeXt-Tiny; 224×224 inputs; batch 48; AdamW, learning rate 5×10^{-5} and weight decay 0.01; bf16; EMA 0.999; at most 50 epochs	Validation macro-F1 with patience 10; mean of five probability vectors
Stage 2a exposure	Five-fold tile-blocked out-of-fold type predictions for training rows	Standardized ridge baseline ($\alpha = 100$) plus absolute-error histogram gradient boosting (350 iterations) on $\log(1 + \hat{p})$	Validation log MAE; nonnegative inverse-log transform
Stage 2b context comparison	Tile-blocked 15% validation split; five matched footprint-only and footprint-plus-ring pairs; 256×256 crops	Cumulative binary cross-entropy; shared ConvNeXt-Tiny; natural sampling; batch 16; AdamW, learning rate 5×10^{-5} and weight decay 0.10; cosine schedule; bf16; EMA 0.999; at most 20 epochs	Validation macro-F1 with QWK tie-break; within-seed comparison of pooling regions
Stage 2b inference	Three independently fitted footprint-plus-ring models with source-tile blocking	Cumulative binary cross-entropy; mild weighted sampling; batch 16; AdamW, learning rate 5×10^{-5} and weight decay 0.05; cosine schedule; bf16; EMA 0.999; at most 20 epochs	Validation macro-F1 with QWK tie-break; 4:3:2 weighted mean of raw cumulative logits

to pre-event crops, post-event crops, and masks. Mild colour jitter is applied independently to the paired RGB observations. The context comparison holds natural sampling and weight decay 0.10 constant across each footprint-only and footprint-plus-ring pair. The operational ensemble instead uses three independently fitted footprint-plus-ring models with mild weighted sampling and weight decay 0.05. In both protocols, checkpoints are selected by validation macro-F1, with QWK resolving ties; post-hoc temperature scaling is fitted on validation predictions for confidence reporting.

Multi-date synthesis. The Los Angeles application applies the flood-trained damage model without hazard-specific retraining. Five post-event dates are aggregated with the valid-coverage rule in Section 1.7. We report the resulting class and unknown-state counts, spatial coverage, and the review fields exposed at the building level.

2.3 Component Comparisons and Sensitivity Analysis

Stage 2a comparisons test the information carried by the footprint mask, geometric vector, imbalance controls, and soft type probabilities. Type-input ablations hold the common split and training seed fixed. The population comparison removes soft type information while retaining the same ridge-plus-residual architecture, thereby isolating the type contribution. Two alternative tile-blocked partitions assess geographic sensitivity, while source- and class-stratified metrics assess dependence on target construction. Probability scaling is evaluated both for type calibration and for its downstream effect on exposure reconstruction.

Stage 2b comparisons isolate neighbourhood context using five matched footprint-only and footprint-plus-ring pairs. Each pair shares its tile partition and training seed and uses natural sampling, so the within-pair difference reflects the pooling region rather than sampling policy. Separate five-seed comparisons assess mild weighted sampling. Calibration is evaluated after model fitting and does not redefine the cumulative ordinal target or the class-selection rule.

2.4 Evaluation Measures

Building delineation is measured by precision, recall, and F1 after IoU-0.5 matching; mean IoU is calculated over matched pairs. Building-type evaluation uses macro-F1, accuracy, and expected calibration error (ECE) (Guo et al., 2017). Exposure error is the mean absolute difference in $\log(1 + p)$, and the factor-two rate is the fraction satisfying

$$|\log(1 + \hat{p}) - \log(1 + \tilde{p})| \leq \log 2. \quad (10)$$

Raw mean absolute error is also reported in proxy-occupancy units. Ordinal damage is evaluated by macro-F1, quadratic weighted kappa (QWK) (Cohen, 1968), ECE, and negative log-likelihood (NLL). All exposure metrics quantify reconstruction of $\tilde{p}$ as defined in Section 1.3.

3 Results

3.1 Building Delineation Establishes the Observable Population

Building discovery, rather than boundary fit among detections, was the principal coverage constraint. Across 933 xView2 images, the evaluation contained 54,862 reference buildings and 22,824 predictions. At IoU 0.5, 15,567 predictions were matched, yielding 0.6820 precision, 0.2837 recall, and 0.4008 F1 (Table 5). The mean IoU of 0.7589 among matched pairs indicates that detected instances were generally well localized, but the low recall restricts which buildings can receive later exposure and damage attributes.

Table 5: Building-instance delineation on the xView2 test set.

Metric	Value
Precision	0.6820
Recall	0.2837
F1	0.4008
Mean IoU, matched pairs	0.7589

3.2 Building Type Supports Exposure-Proxy Reconstruction

The type and exposure models were stable across five trainings on the common tile-blocked partition. Mean type macro-F1 was 0.8529 ± 0.0106 , while exposure log MAE was 0.0760 ± 0.0032 and the factor-two rate was 0.9603 ± 0.0034 (Table 6). Averaging type probabilities increased macro-F1 to 0.857898 and reduced exposure log MAE to 0.073351. The ensemble therefore improved both the intermediate type representation and reconstruction of the rule-derived exposure target.

Soft type information accounted for most of the regressor’s measured gain. With the residual model held fixed, removing predicted type increased log MAE from 0.0742 to 0.1157; a direct gradient-boosting model without type reached 0.1175. In the type-model ablation, RGB alone produced 0.8048 macro-F1, adding the footprint mask raised it to 0.8626, adding geometric attributes produced 0.8601, and removing imbalance handling reduced it to 0.8262. The footprint mask and imbalance treatment therefore contributed more than the additional geometry vector in this comparison.

Table 6: Building-type and exposure-proxy results on the common tile-blocked partition. Raw MAE is in proxy-occupancy units.

Model summary	Type macro-F1	Accuracy	ECE	Log MAE	Factor-two	Raw MAE
Five-run stability	0.8529 ± 0.0106	0.9300 ± 0.0035	0.0542 ± 0.0042	0.0760 ± 0.0032	0.9603 ± 0.0034	14.76 ± 0.91
Probability ensemble	0.857898	0.933208	0.034243	0.073351	0.960674	14.3529

Table 7: Exposure-proxy error by label-construction source and building type.

Slice	n	Log MAE	Factor-two
Area rule	2,815	0.027934	0.987922
Parcel code	389	0.402013	0.763496
Commercial	2,626	0.029157	0.984768
Residential multi	513	0.276811	0.847953
Residential small	65	0.253009	0.876923

3.3 Exposure Error Reflects Target Source and Spatial Partition

Aggregate exposure accuracy concealed marked source and class differences. Log MAE was 0.027934 for area-rule labels and 0.402013 for parcel-code labels, a 14.4-fold difference (Table 7). Commercial buildings, which dominate the validation data, had substantially lower error than either residential class. These results connect model behavior to how the proxy was geographically constructed, rather than treating the aggregate error as uniform across buildings.

Performance also varied with the geographic partition. Across the reference and two alternative tile assignments, type macro-F1 ranged from 0.842816 to 0.890720 and exposure log MAE from 0.061295 to 0.104110 (Table 8). Reporting this variation alongside source-stratified error identifies where the proxy is reproduced reliably and where interpretation should be more cautious.

Probability scaling further illustrates the coupling between pipeline components. Temperature and classwise scaling reduced type ECE from 0.03424 to 0.00972 and 0.00644, respectively, but increased downstream exposure log MAE from 0.07335 to 0.08267 and 0.08159. Raw ensemble probabilities were therefore retained for exposure estimation: better standalone calibration did not improve the quantity ultimately delivered by this branch.

3.4 Neighbourhood Context Improves Ordinal Damage Estimation

Adding the context ring improved macro-F1 in every matched comparison (Table 9). Across five joint partition/training seeds, mean macro-F1 increased from 0.6358 ± 0.0299 to 0.6781 ± 0.0262 , and mean QWK increased from 0.7138 ± 0.0291 to 0.8309 ± 0.0215 . The paired macro-F1 gain was 0.0423 ± 0.0107 . The repeated increase shows that neighbourhood change contributed information beyond the target footprint alone.

The matched architecture comparison used natural sampling to isolate the context effect. In separate five-seed fits of the context model, mild weighted sampling increased mean macro-F1 from 0.6781 to 0.6868, while mean QWK decreased from 0.8309 to 0.8264 and ECE increased from 0.0434 to 0.0621. For the selected anchor checkpoint, temperature scaling reduced ECE from 0.057250 to 0.035605 without changing ordinal classes. Calibrated probabilities were therefore used for confidence indicators, while raw cumulative logits determined the damage class.

Table 8: Sensitivity to tile-blocked training and validation partitions.

Partition	Type	macro-F1	Log MAE	Factor-two
Reference		0.857898	0.073351	0.960674
Alternative A		0.842816	0.104110	0.944844
Alternative B		0.890720	0.061295	0.972550

Table 9: Matched comparisons of footprint-only (M) and footprint-plus-ring ($M+R$) pooling under natural sampling.

Seed	M F1	M QWK	M ECE	$M+R$ F1	$M+R$ QWK	$M+R$ ECE
42	0.6064	0.6936	0.0961	0.6633	0.8203	0.0376
123	0.6198	0.7261	0.0601	0.6695	0.8466	0.0425
3407	0.6179	0.6894	0.0613	0.6487	0.8048	0.0403
2025	0.6759	0.7007	0.0521	0.7144	0.8242	0.0628
7777	0.6588	0.7592	0.0623	0.6944	0.8584	0.0340
Mean $\pm$ sd	0.6358 ± 0.0299	0.7138 ± 0.0291	0.0664 ± 0.0171	0.6781 ± 0.0262	0.8309 ± 0.0215	0.0434 ± 0.0113

3.5 Coverage-Aware Synthesis Preserves Unknown States

The Los Angeles application translated repeated predictions into building-level records without converting absent imagery into apparent no damage. The five-date summary contained 21,797 delineated buildings across 120 of 295 spatial cells. Coverage-aware voting assigned 1,245 buildings to the explicit not-identifiable class, alongside 16,246 no-visible-damage, 4,174 minor, 39 major, and 93 destroyed predictions. Thus 5.7% of mapped buildings remained visibly outside the model’s valid observational support instead of entering the lowest damage class.

The synthesis table carries valid-date counts, class probabilities, uncertainty indicators, and source references alongside each damage estimate. Analysts can therefore select severe buildings, inspect cases with high uncertainty indicators, or separate unobserved buildings before allocating attention. Figure 5 connects these fields back to imagery and footprint geometry, from an individual building to the regional synthesis. The practical contribution is this traceable transition from separate GeoAI outputs to reviewable instance records: coverage, proxy exposure, ordinal damage, and uncertainty indicators remain distinct attributes rather than being hidden inside one composite prediction.

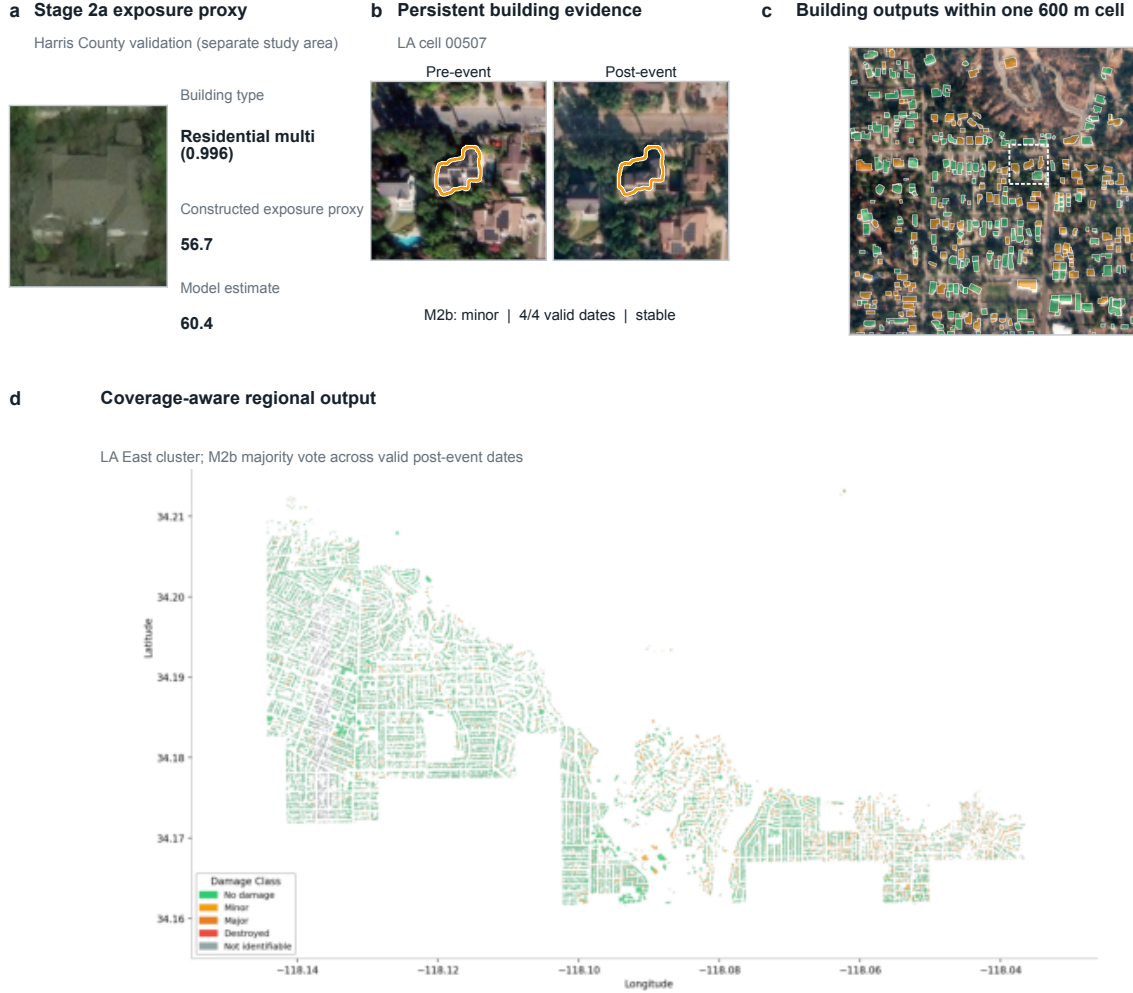


Figure 5: Building-level evidence and coverage-aware spatial delivery. (a) A representative Harris County validation record shows the Stage 2a type prediction, constructed exposure proxy, and model estimate in proxy-occupancy units; this panel is from a separate study area from the Los Angeles application. (b) A persistent Los Angeles building is linked across pre- and post-event imagery and retains its coverage-aware multi-date (M2b) damage record. (c) M2b outputs for the surrounding 600-m cell. (d) M2b synthesis for the eastern Los Angeles study area, with not-identifiable buildings retained separately from no visible damage. The flood-trained damage model was applied without wildfire-specific retraining, so panels (b)–(d) demonstrate traceable spatial delivery and coverage handling rather than validated wildfire-damage accuracy.

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